

Informe RF-001

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Requerimiento

Para este trabajo nos pidieron implementar el **RF-001**: “El sistema debe permitir contar la cantidad de elementos por categoria, el mismo que debe imprimir en la barra de tareas”.

Por ejemplo, en el screen se puede notar que existen 2 Package, 2 Class y 1 Component, y el contador deberia mostrarlos en la barra inferior:

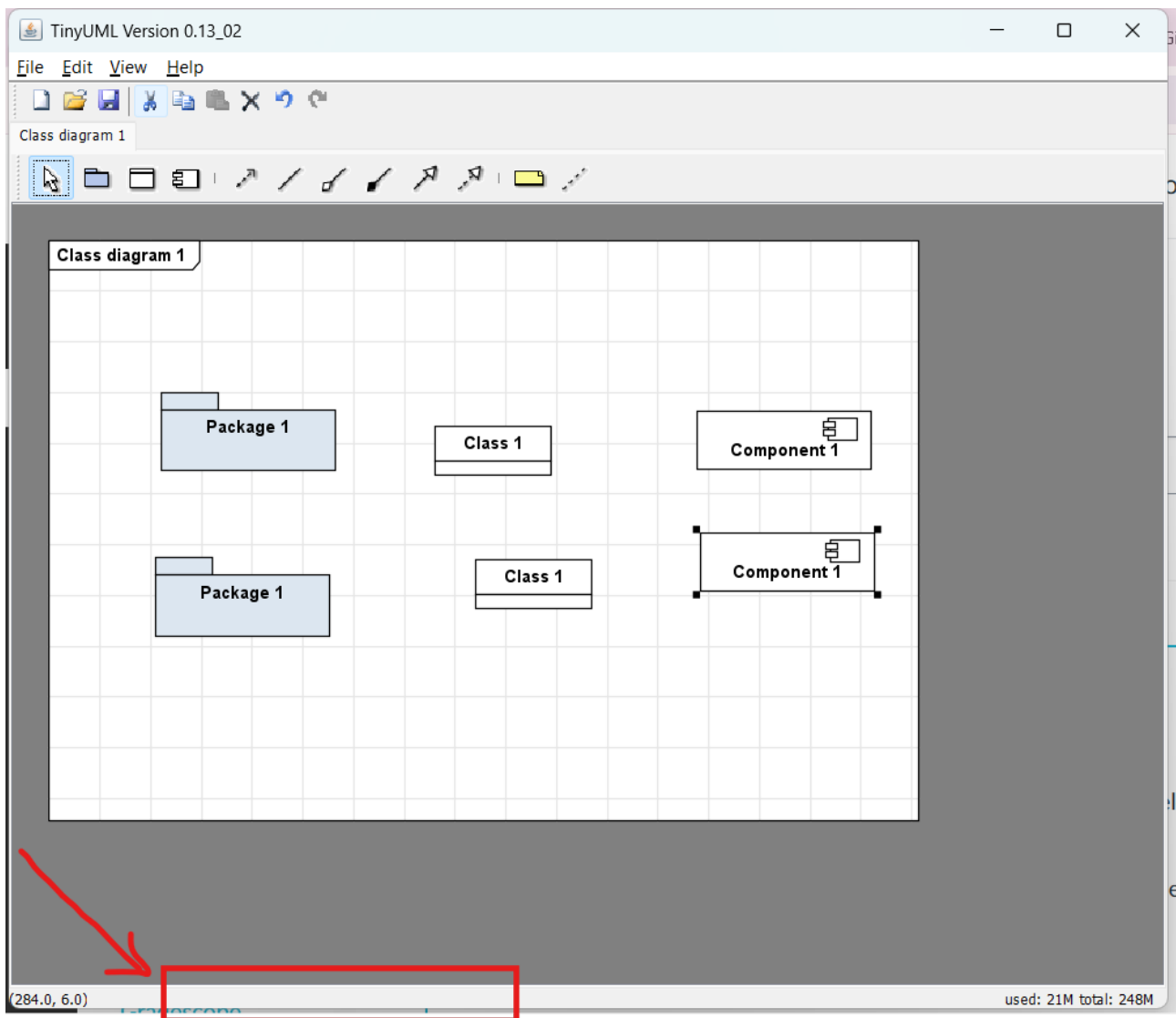


Figure 1: Ejemplo del contador en la barra de tareas

Archivos modificados

Para que esto funcione tuvimos que tocar tres archivos. Acá se explica que se hizo en cada uno y por que.

1. AppFrame.java

Acá es donde se implemento el contador. Agregamos dos metodos nuevos:

- `updateElementCount()`: arma el texto de la barra de tareas con el total y el detalle por categoria (Package, Class, Component, Note). Si una categoria tiene cero elementos no la imprime para que no quede ruido en la barra.
- `countElements(CompositeNode parent, int[] counts)`: recorre de forma recursiva los nodos del diagrama, contando por tipo. Ignora las Connection (asociaciones, dependencias, herencia, conexiones de notas) porque solo nos interesan los elementos visuales como nodos. Es recursivo para que los elementos dentro de un Package tambien se cuenten.

Tambien metimos el `countLabel` en el centro del statusbar para que el contador sea visible.

```
tinyuml-0.13_02 > src > main > java > org > tinyuml > ui > AppFrame.java
47 import javax.swing.JFileChooser;
48 import javax.swing.filechooser.FileNameExtensionFilter;
49 import org.tinyuml.draw.Label;
50 import org.tinyuml.draw.CompositeNode;
51 import org.tinyuml.draw.Connection;
52 import org.tinyuml.draw.DiagramElement;
53 import org.tinyuml.draw.LabelChangeListener;
54 import org.tinyuml.model.UmlModel;
55 import org.tinyuml.util.AppCommandListener;
56 import org.tinyuml.uml.draw.shared.NoteElement;
57 import org.tinyuml.uml.draw.structure.ClassElement;
58 import org.tinyuml.uml.draw.structure.ComponentElement;
59 import org.tinyuml.uml.draw.structure.PackageElement;
60 import org.tinyuml.uml.draw.structure.StructureDiagram;

83 public class AppFrame extends JFrame
84 implements EditorStateListener, AppCommandListener, SelectionListener {
85
86     private JTabbedPane tabbedPane;
87     private JLabel coordLabel = new JLabel(" ");
88     private JLabel countLabel = new JLabel("Total Items:00");
89     private JLabel memLabel = new JLabel(" ");
90     private UmlModel umlModel;
91     private DiagramEditor currentEditor;
92     private transient Timer timer = new Timer();
93     private transient StaticStructureEditorToolBarManager staticToolBarManager;
94     private transient EditorCommandDispatcher editorDispatcher;
95     private transient MainToolBarManager toolBarManager;
96     private transient MenuManager menuManager;
97     private transient File currentFile;
98     private LinkedList<DiagramEditor> diagramEditors;
99     private Collection<DiagramElement> lastCopiedElements;
100     private transient Map<String, MethodCall> selectorMap =
101         new HashMap<String, MethodCall>();
```

```

200 private void createEditor(StructureDiagram diagram) {
201     currentEditor = new DiagramEditor(this, diagram);
202     currentEditor.addEditorStateListener(this);
203     currentEditor.addSelectionListener(this);
204     currentEditor.addAppCommandListener(editorDispatcher);
205     currentEditor.addAppCommandListener(this);
206     JScrollPane spane = new JScrollPane(currentEditor);
207     JPanel editorPanel = new JPanel(new BorderLayout());
208     spane.getVerticalScrollBar().setUnitIncrement(10);
209     spane.getHorizontalScrollBar().setUnitIncrement(10);
210     staticToolBarManager = new StaticStructureEditorToolBarManager();
211     JToolBar toolbar = staticToolBarManager.getToolBar();
212     staticToolBarManager.addCommandListener(editorDispatcher);
213     editorPanel.add(spane, BorderLayout.CENTER);
214     editorPanel.add(toolbar, BorderLayout.NORTH);
215     final Component comp = tabbedPane.add(diagram.getLabelText(), editorPanel);
216     diagram.addNameLabelChangeListener(new LabelChangeListener() {
217         /** {@inheritDoc} */
218         public void labelTextChanged(Label label) {
219             tabbedPane.setTitleAt(tabbedPane.indexOfComponent(comp),
220                 label.getText());
221         }
222     });
223     updateElementCount();
224 }

```

```

66 /**
67  * RF-001: Refreshes the status bar element counter for the current diagram.
68  * Counts nodes by category (Package, Class, Component, Note) and ignores
69  * connections (associations, dependencies, inheritance, note connections).
70  */
71 private void updateElementCount() {
72     int[] counts = new int[4]; // 0=Package, 1=Class, 2=Component, 3=Note
73     if (currentEditor != null && currentEditor.getDiagram() != null) {
74         countElements(currentEditor.getDiagram(), counts);
75     }
76     int total = counts[0] + counts[1] + counts[2] + counts[3];
77     StringBuilder sb = new StringBuilder();
78     sb.append(String.format("Total Items:%02d", total));
79     if (counts[0] > 0) sb.append(String.format("; Package:%02d", counts[0]));
80     if (counts[1] > 0) sb.append(String.format("; Class:%02d", counts[1]));
81     if (counts[2] > 0) sb.append(String.format("; Component:%02d", counts[2]));
82     if (counts[3] > 0) sb.append(String.format("; Note:%02d", counts[3]));
83     countLabel.setText(sb.toString());
84 }

```

```

276
277 /**
278  * Recursively counts diagram nodes by category. Walks composite nodes so
279  * elements nested inside packages are also counted.
280  */
281 private void countElements(CompositeNode parent, int[] counts) {
282     for (DiagramElement child : parent.getChildren()) {
283         if (child instanceof Connection) continue;
284         if (child instanceof PackageElement) counts[0]++;
285         else if (child instanceof ClassElement) counts[1]++;
286         else if (child instanceof ComponentElement) counts[2]++;
287         else if (child instanceof NoteElement) counts[3]++;
288         if (child instanceof CompositeNode) {
289             countElements((CompositeNode) child, counts);
290         }
291     }
292 }
293
687 public void paste(){
688     getCurrentEditor().pasteElement(lastCopiedElements);
689     updateElementCount();
690 }

```

2. CompositeNode.java

Cuando intentamos compilar fallaba en “cannot find symbol: method getChildren()”. El metodo getChildren() si existe en la clase concreta **AbstractCompositeNode**, pero no estaba declarado en el interfaz **CompositeNode**. Como se accedia al objeto a traves del tipo del interfaz, el compilador no veia el metodo.

La solucion fue declarar **getChildren()** dentro del interfaz. No tuvimos que tocar la implementacion porque ya esta hecha en **AbstractCompositeNode**, solo se expuso.

```

tinyuml-0.13_02 > src > main > java > org > tinyuml > draw > CompositeNode.java
1 /**
17  * along with TinyUML; if not, write to the Free Software
18  * Foundation, Inc., 51 Franklin St, Fifth Floor, Boston, MA 02110-1301 USA
19  */
20 package org.tinyuml.draw;
21
22 import java.awt.geom.Rectangle2D;
23 import java.util.Collection;
24 import java.util.List;
25
26 /**
27  * The base interface for Nodes that can contain other children. To allow for
28  * separation of leaf nodes and non-leaf nodes, a specific composite node
29  * is introduced.
30  * @author Wei-ju Wu
31  * @version 1.0
32  */
33 public interface CompositeNode extends Node, CompositeElement {
34
35     /**
36      * Returns the list of child elements contained in this node.
37      * @return the children
38      */
39     List<DiagramElement> getChildren();
40

```

Figure 2: Cambios en CompositeNode.java

3. pom.xml

Al compilar tambien salian errores de tipo “unmappable character for encoding UTF8” en varios archivos del proyecto original. Esto pasa porque los .java del proyecto son del 2007 y estan guardados en Latin-1 (tienen acentos en los comentarios), pero el Maven moderno asume UTF-8 por defecto.

En vez de andar convirtiendo archivo por archivo, se agrego un bloque de properties en el pom.xml para decirle a Maven que los fuentes estan en ISO-8859-1. Asi no toque ningun .java por este motivo.



```
tinymuml-0.13_02 > pom.xml
You, 21 minutes ago | 2 authors (tchambil and one other)
1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-1
2   xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v
3   <modelVersion>4.0.0</modelVersion>
4   <groupId>org.tinymuml</groupId>
5   <artifactId>tinymuml</artifactId>
6   <packaging>jar</packaging>
7   <version>0.13_02</version>
8
9   <properties>
10    <project.build.sourceEncoding>ISO-8859-1</project.build.sourceEncoding>
11  </properties>
```

Figure 3: Cambios en pom.xml

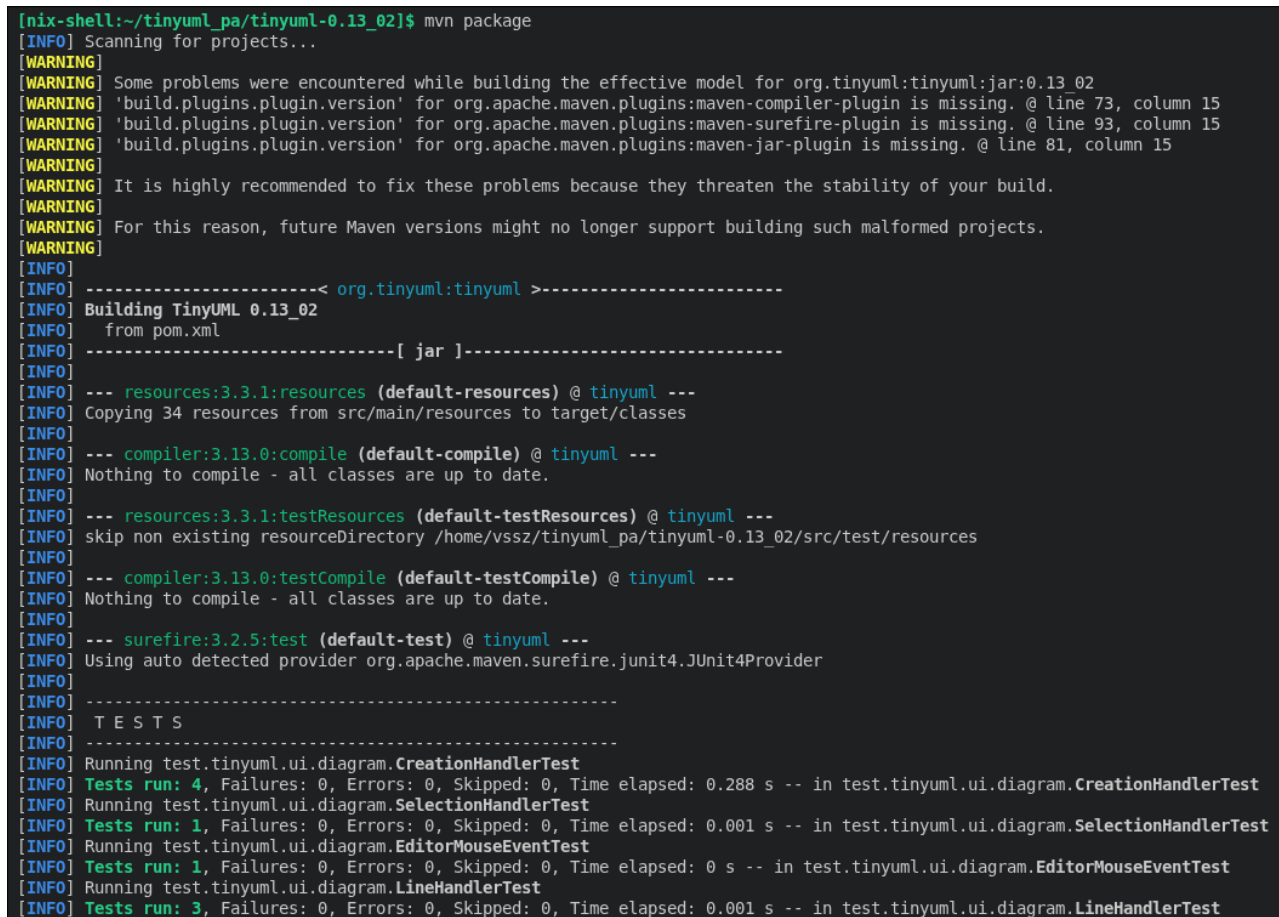
Como correr el proyecto

Estoy en NixOS, asi que primero entro a un shell con JDK y Maven:

```
nix-shell -p jdk8 maven
```

Compilo:

```
mvn package
```



```
[nix-shell:~/tinymuml_pa/tinymuml-0.13_02]$ mvn package
[INFO] Scanning for projects...
[WARNING]
[WARNING] Some problems were encountered while building the effective model for org.tinymuml:tinymuml:jar:0.13_02
[WARNING] 'build.plugins.plugin.version' for org.apache.maven.plugins:maven-compiler-plugin is missing. @ line 73, column 15
[WARNING] 'build.plugins.plugin.version' for org.apache.maven.plugins:maven-surefire-plugin is missing. @ line 93, column 15
[WARNING] 'build.plugins.plugin.version' for org.apache.maven.plugins:maven-jar-plugin is missing. @ line 81, column 15
[WARNING]
[WARNING] It is highly recommended to fix these problems because they threaten the stability of your build.
[WARNING]
[WARNING] For this reason, future Maven versions might no longer support building such malformed projects.
[WARNING]
[INFO]
[INFO] -----< org.tinymuml:tinymuml >-----
[INFO] Building TinyUML 0.13_02
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ tinymuml ---
[INFO] Copying 34 resources from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ tinymuml ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ tinymuml ---
[INFO] skip non existing resourceDirectory /home/vssz/tinymuml_pa/tinymuml-0.13_02/src/test/resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ tinymuml ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ tinymuml ---
[INFO] Using auto detected provider org.apache.maven.surefire.junit4.JUnit4Provider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running test.tinymuml.ui.diagram.CreationHandlerTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.288 s -- in test.tinymuml.ui.diagram.CreationHandlerTest
[INFO] Running test.tinymuml.ui.diagram.SelectionHandlerTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in test.tinymuml.ui.diagram.SelectionHandlerTest
[INFO] Running test.tinymuml.ui.diagram.EditorMouseEventTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinymuml.ui.diagram.EditorMouseEventTest
[INFO] Running test.tinymuml.ui.diagram.LineHandlerTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in test.tinymuml.ui.diagram.LineHandlerTest
```

```

[INFO] Running test.tinyuml.umldraw.structure.AssociationTest
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.008 s -- in test.tinyuml.umldraw.structure.AssociationTest
[INFO] Running test.tinyuml.umldraw.structure.PackageElementTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.umldraw.structure.PackageElementTest
[INFO] Running test.tinyuml.umldraw.shared.BaseConnectionTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.004 s -- in test.tinyuml.umldraw.shared.BaseConnectionTest
[INFO] Running test.tinyuml.umldraw.shared.NoteElementTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.umldraw.shared.NoteElementTest
[INFO] Running test.tinyuml.umldraw.shared.UmlConnectionSelectionTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.umldraw.shared.UmlConnectionSelectionTest
[INFO] Running test.tinyuml.umldraw.shared.NamedElementLabelSourceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in test.tinyuml.umldraw.shared.NamedElementLabelSourceTest
[INFO] Running test.tinyuml.util.ApplicationResourcesTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.util.ApplicationResourcesTest
[INFO] Running test.tinyuml.util.MethodCallTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.util.MethodCallTest
[INFO] Running test.tinyuml.model.UmlPackageTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.model.UmlPackageTest
[INFO] Running test.tinyuml.model.UmlRelationTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s -- in test.tinyuml.model.UmlRelationTest
[INFO] Running test.tinyuml.model.UmlPropertyTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.model.UmlPropertyTest
[INFO] Running test.tinyuml.model.UmlClassTest
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.model.UmlClassTest
[INFO] Running test.tinyuml.model.UmlModelElementTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.model.UmlModelElementTest
[INFO] Running test.tinyuml.model.DefaultNamedElementTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.model.DefaultNamedElementTest
[INFO] Running test.tinyuml.model.MultiplicityTest
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.model.MultiplicityTest
[INFO] Running test.tinyuml.model.UmlModelTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.model.UmlModelTest
[INFO] Running test.tinyuml.model.UmlStereotypeTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0 s -- in test.tinyuml.model.UmlStereotypeTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 278, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ tinyuml ---
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.156 s
[INFO] Finished at: 2026-05-26T00:34:06-05:00
[INFO] -----

```

Lanzo el programa:

```
mvn exec:java -Dexec.mainClass=org.tinyuml.Main
```

Y al agregar elementos al diagrama, la barra inferior se actualiza con el conteo por categoria:

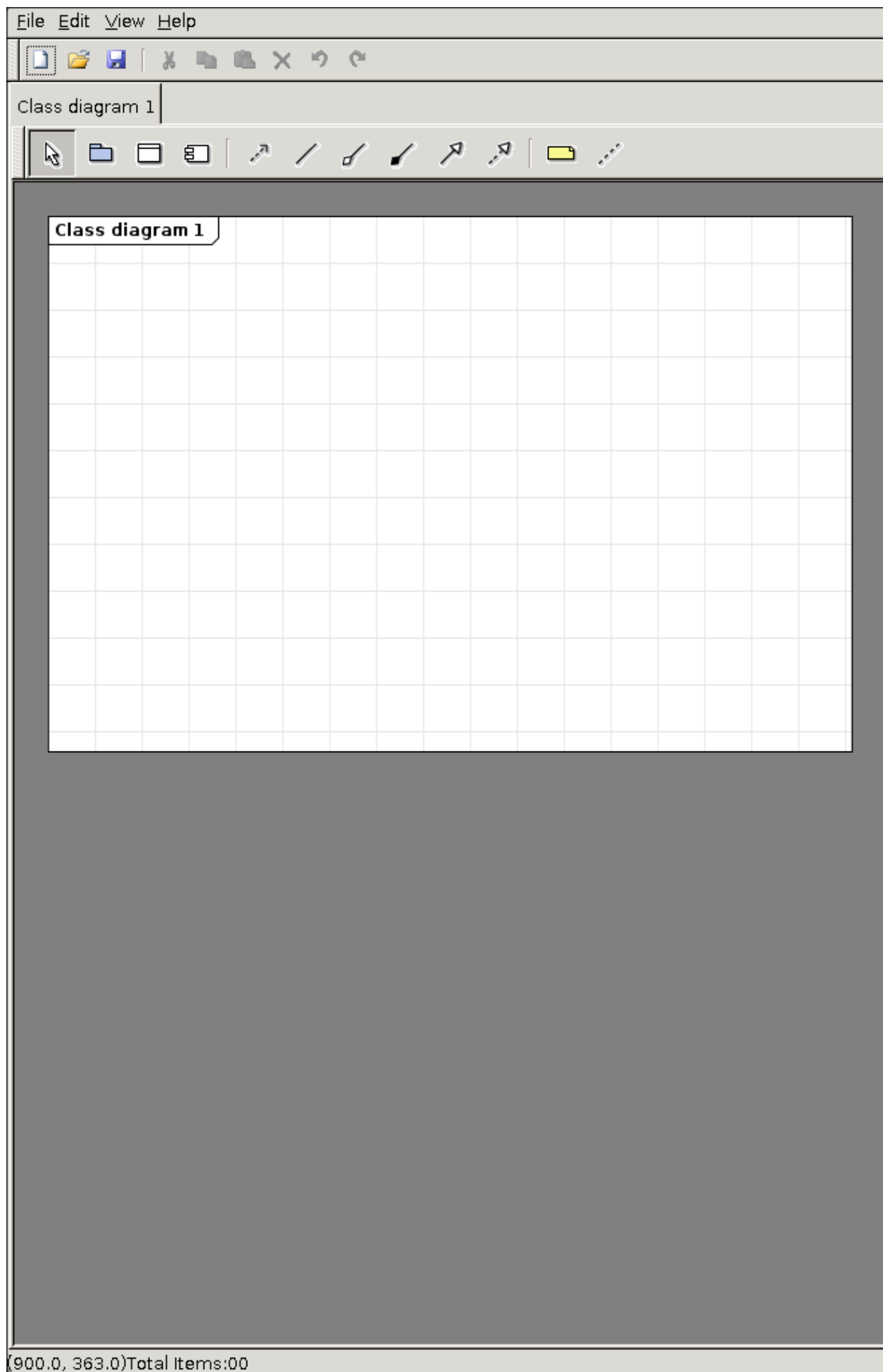


Figure 4: Aplicacion abierta sin elementos

